

MLA0301– REINFORCEMENT LEARNING

ASSIGNMENT

A. Assignment Information

Department:	AI&ML
Programme:	FACULTY OF ENGINEERING
Course Code & Course Name	MLA0301 & REINFORCEMENT LEARNING
Academic Year / Batch	2026-2027
Faculty Name	Dr.G. A. SENTHIL
Assignment Title	Analyse and design an RL-based patient health monitoring and treatment recommendation system using Markov Decision Process and Bellman equations. Model patient health conditions as states, treatment interventions as actions, and clinical outcomes as rewards, and implement an optimal policy to recommend appropriate treatment strategies based on changing patient conditions. Analyse the learned policy using treatment effectiveness, recovery rate, and cumulative reward.
Date of Issue	01-09-2026
Date of Submission	02-09-2026
Maximum Marks	100
Course Outcome(s) - CO	CO1-Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems. CO2 - Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies. CO3 - Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. CO4-Evaluate model-based reinforcement learning approaches for prediction, planning, and policy optimization. CO5-Apply hierarchical, meta-learning, and multi-agent RL approaches to solve complex decision-making problems.

Bloom's Taxonomy Level	BL3 (Apply) - <i>Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks.</i> BL4 (Analyse) - <i>Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments.</i> BL5 (Evaluate) - <i>Model-based reinforcement learning approaches for prediction, planning, and policy optimization.</i> BL6 (Create) - <i>Develop an MDP-based solution for an intelligent traffic signal system by defining the state space, action space, reward function, transition model, and optimal policy.</i>
SDG Mapping (SDG 1-17)	SDG 3 – (Good Health and Well-Being) Enables intelligent patient health monitoring and treatment recommendation to support timely clinical decision-making, improve patient recovery, and promote better healthcare outcomes. SDG 9 – (Industry, Innovation & Infrastructure) Applies Markov Decision Processes, Bellman equations, and Reinforcement Learning to develop innovative AI-based healthcare monitoring and treatment-support systems. SDG 10 – (Reduced Inequalities) Supports consistent and data-driven treatment recommendations that can assist healthcare providers in delivering appropriate care across different patient conditions and populations. SDG 12 – (Responsible Consumption & Production) Optimizes treatment decisions and healthcare resources by selecting effective interventions based on changing patient health conditions, helping reduce unnecessary treatments and resource usage. SDG 16 – (Peace, Justice & Strong Institutions) Promotes reliable, systematic, and transparent decision-making in healthcare through an explainable RL-based policy for treatment recommendations.
Industry / Societal Relevance	The assignment addresses the growing need for AI-driven healthcare decision support by developing a Reinforcement Learning-based patient health monitoring and treatment recommendation system. The solution can help healthcare professionals analyze changing patient health conditions, select appropriate treatment interventions, and optimize clinical outcomes using Markov Decision Processes and Bellman equations. Industry-wise, it supports intelligent healthcare monitoring, personalized treatment planning, and efficient clinical decision-making. Societally, it can contribute to better patient care, improved recovery rates, timely treatment, reduced healthcare resource wastage, and accessible data-driven healthcare services.

Question:

Analyse and design an RL-based patient health monitoring and treatment recommendation system using Markov Decision Process and Bellman equations. Model patient health conditions as states, treatment interventions as actions, and clinical outcomes as rewards, and implement an optimal policy to recommend appropriate treatment strategies based on changing patient conditions. Analyse the learned policy using treatment effectiveness, recovery rate, and cumulative reward.

Answer:**1. PROBLEM STATEMENT AND PROBLEM FORMULATION****1.1 Problem Statement**

Patient health conditions can change over time, so continuous monitoring and appropriate intervention are important. Traditional systems mainly use fixed rules and thresholds, which may not adapt effectively to changing conditions.

This assignment proposes an **RL-based Patient Health Monitoring and Treatment Recommendation System** using a **Markov Decision Process (MDP)**. Patient conditions are represented as states, possible interventions as actions, and clinical outcomes as rewards. The RL agent learns an optimal policy to select suitable actions based on the patient's changing condition.

The system is designed as an **educational simulation** and not as a real clinical treatment system.

1.2 Problem Formulation

The problem is formulated as an MDP:

$$\text{MDP} = (\mathbf{S}, \mathbf{A}, \mathbf{P}, \mathbf{R}, \gamma)$$

where:

- **S** – Patient health states
- **A** – Available actions
- **P** – State transition probabilities
- **R** – Reward function

- γ – Discount factor

The objective is to learn a policy that maximizes the expected cumulative reward and improves the simulated patient outcome.

1.3 State Space

State	Description
Stable	Patient condition is normal
Mild	Mild health abnormality
Moderate	Requires closer monitoring
Severe	Requires urgent attention
Recovery	Patient condition is improving

1.4 Action Space

Action	Description
Monitoring	Continue monitoring
Medication	Simulate medication intervention
Increased Monitoring	Monitor more frequently
Hospital Referral	Recommend professional medical care

1.5 Reward Function

Outcome	Reward
Recovery	+10
Improvement	+5
Stable condition	+2
No change	0

Worsening	-5
Severe deterioration	-10

Positive rewards encourage effective actions, while negative rewards discourage actions associated with worsening conditions.

1.6 Bellman Equation

The Bellman equation helps the RL agent determine the long-term value of an action:

$$V(s) = \max [R(s,a) + \gamma V(s')]$$

where $V(s)$ is the value of the current state, $R(s,a)$ is the immediate reward, γ is the discount factor, and $V(s')$ is the value of the next state.

2. OBJECTIVE AND EXPECTED OUTCOMES

2.1 Objectives

The main objectives of the proposed system are:

- To develop an RL-based patient health monitoring system.
- To model patient health conditions using a **Markov Decision Process**.
- To select suitable treatment or monitoring actions based on the patient's condition.
- To use the **Bellman equation** to learn an optimal policy.
- To evaluate the learned policy using treatment effectiveness, recovery rate, and cumulative reward.

2.2 Expected Outcomes

The expected outcomes of the system are:

- An MDP model representing different patient health conditions.
- An RL agent that learns suitable actions for each state.
- An optimal policy for treatment recommendation.
- Improved simulated patient recovery through appropriate actions.
- Measurement of **treatment effectiveness, recovery rate, and cumulative reward**.

- A simple and explainable demonstration of RL application in healthcare decision support.

3. REQUIREMENTS, CONSTRAINTS AND ASSUMPTIONS

3.1 Software Requirements

- Python 3.x
- Jupyter Notebook or VS Code
- NumPy
- Pandas
- Matplotlib
- Scikit-learn (optional)
- Operating System: Windows/Linux

3.2 Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Basic processor
- Internet connection for installing required libraries

3.3 Functional Requirements

The system should:

- Accept the patient's current health state.
- Select an appropriate action using the learned RL policy.
- Calculate the reward for the selected action.
- Update the patient's next health state.
- Learn the optimal policy through repeated interactions.
- Calculate recovery rate and cumulative reward.

- Display the final learned policy and results.

3.4 Constraints

- The system uses a **simulated patient environment**.
- Patient conditions are represented using a limited number of discrete states.
- Treatment actions are represented as general categories.
- The transition probabilities and rewards are predefined for the simulation.
- The system is not intended for real-world medical diagnosis or prescription.

3.5 Assumptions

- The patient's current condition can be represented by one of the defined states.
- Each action can produce a measurable simulated outcome.
- Rewards correctly represent improvement or deterioration.
- The patient state satisfies the Markov property, where the next state mainly depends on the current state and selected action.

4. APPLICATION OF RELEVANT COURSE KNOWLEDGE / CONCEPTS

The proposed system applies important **Reinforcement Learning** concepts to patient health monitoring and treatment recommendation.

4.1 Reinforcement Learning

Reinforcement Learning is used to train an agent to select suitable actions by learning from rewards and penalties.

4.2 Markov Decision Process

The patient monitoring problem is represented as an MDP consisting of **states, actions, transitions, rewards, and discount factor**.

4.3 State

A state represents the current health condition of the patient, such as **Stable, Mild, Moderate, Severe, and Recovery**.

4.4 Action

An action represents the intervention selected by the RL agent, such as **Monitoring, Medication, Increased Monitoring, and Hospital Referral**.

4.5 Reward

Rewards indicate whether the selected action produces a good or poor outcome. Improvement and recovery receive positive rewards, while worsening conditions receive negative rewards.

4.6 Bellman Equation

The Bellman equation helps calculate the expected future value of each state-action decision and supports selection of the optimal action.

4.7 Optimal Policy

The learned policy maps each patient health state to the most suitable action based on the accumulated learning experience.

4.8 Policy Evaluation

The learned policy is evaluated using **treatment effectiveness, recovery rate, and cumulative reward** to determine the performance of the system.

5. DESIGN / PROPOSED SOLUTION / METHODOLOGY

5.1 Proposed System

The proposed system uses **Reinforcement Learning** to monitor simulated patient conditions and recommend suitable actions. The RL agent observes the current patient state, selects an action, receives a reward, and learns from the resulting health condition.

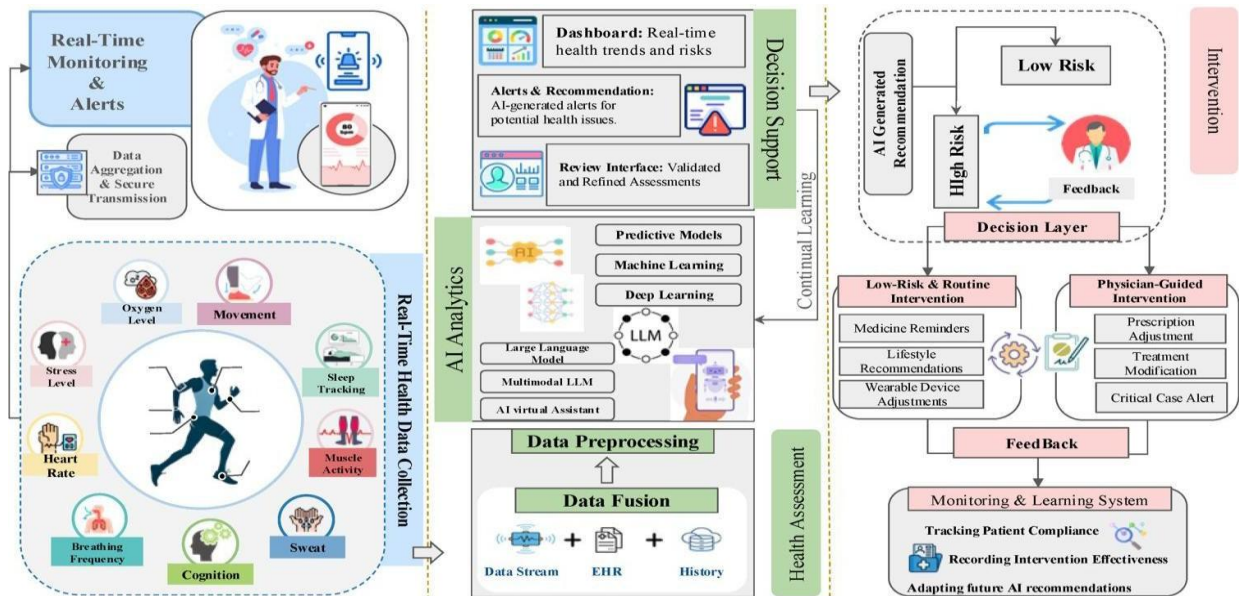
5.2 System Workflow

Patient Condition → Current State → RL Agent → Select Action → Reward → Next State → Policy Update

5.3 Methodology

1. Define patient health states such as Stable, Mild, Moderate, Severe, and Recovery.
2. Define possible actions such as Monitoring, Medication, Increased Monitoring, and Hospital Referral.

3. Assign rewards based on the outcome of each action.
4. Define transition probabilities between patient states.
5. Initialize the value or Q-value table.
6. Apply the Bellman-based learning process to update values.
7. Repeat the process for multiple episodes.
8. Select the action with the highest learned value.
9. Generate the final optimal policy.
10. Evaluate the policy using recovery rate, treatment effectiveness, and cumulative reward.



5.4 System Components

Component	Purpose
Patient State	Represents current health condition
RL Agent	Selects the best action
Action	Represents monitoring or intervention
Reward	Measures the action outcome

Component	Purpose
Environment	Simulates patient condition changes
Policy	Provides the best action for each state
Performance Metrics	Measures system effectiveness

5.5 Expected Policy

After training, the system may learn a policy such as:

Patient State	Recommended Action
Stable	Monitoring
Mild	Medication
Moderate	Increased Monitoring
Severe	Hospital Referral
Recovery	Monitoring

The actual policy is obtained from the learning process rather than being manually fixed.

6. ALGORITHM / PSEUDOCODE / FLOWCHART

6.1 Algorithm

1. Start the system.
2. Define patient health states and available actions.
3. Initialize the Q-value table.
4. Set the learning rate, discount factor, and number of training episodes.
5. Select the current patient state.
6. Select an action using the RL policy.
7. Apply the action to the simulated environment.
8. Observe the next patient state and reward.

9. Update the Q-value using the Bellman equation.
10. Repeat the process until the training episodes are completed.
11. Select the action with the highest Q-value for each state.
12. Generate the optimal policy.
13. Calculate recovery rate, treatment effectiveness, and cumulative reward.
14. Display the results.
15. Stop.

6.2 Pseudocode

START

Define patient states

Define possible actions

Initialize Q-values

Set learning

rate Set

discount factor

Set training episodes

FOR each episode:

 Initialize patient

 state

 WHILE patient is not

 recovered: Select an

 action

 Perform the

 action

 Observe next

 state Calculate

 reward

 Update Q-value using Bellman equation

 Move to next state

END WHILE

END FOR

For each patient state:

 Select action with highest Q-value

Generate optimal policy

Calculate:

 Recovery

 Rate

 Treatment

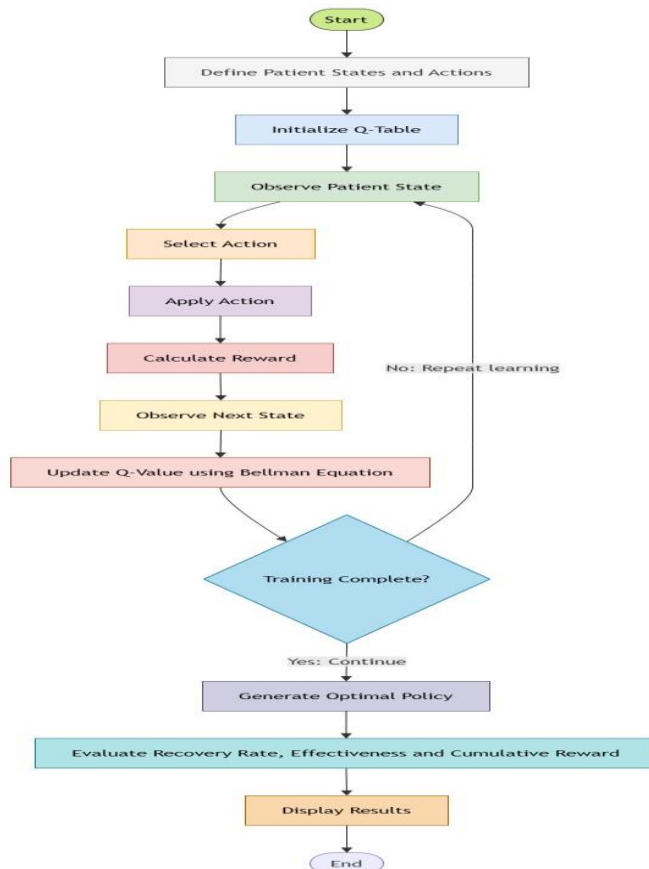
 Effectiveness

 Cumulative Reward

Display results

STOP

6.3 Flowchart



7. IMPLEMENTATION / SOURCE CODE AND ENVIRONMENT / TOOLS USED

7.1 Implementation

The proposed system is implemented using **Python**. A simulated patient environment is created with different health states and treatment actions. The RL agent learns the best action for each state using Q-learning and the Bellman equation.

During training, the agent repeatedly observes the patient state, selects an action, receives a reward, and updates the Q-values. After training, the action with the highest Q-value is selected as the optimal recommendation.

7.2 Environment and Tools

Tool / Technology	Purpose
Python	Main programming language
NumPy	Q-table and numerical calculations
Pandas	Result analysis and tables
Matplotlib	Graphs and visualization
VS Code	Code development and execution
Windows	Operating environment

7.3 System Input

The system takes the **current simulated patient health condition** as input.

Example:

Patient State: Moderate

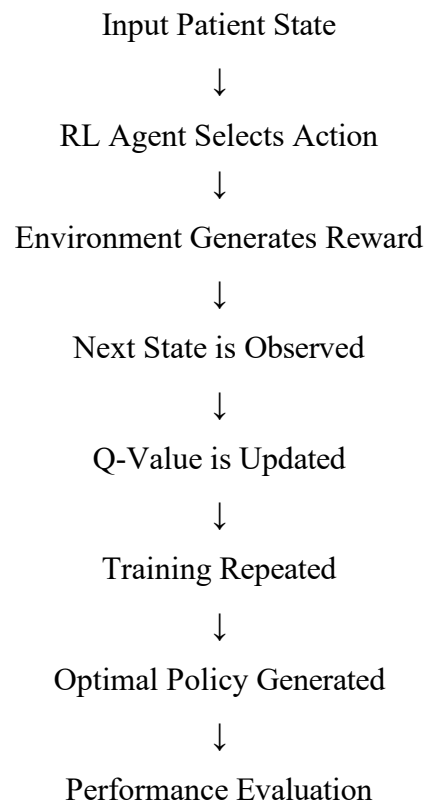
7.4 System Output

The system provides:

- Recommended action
- Next patient state

- Reward received
- Learned Q-values
- Optimal policy
- Recovery rate
- Treatment effectiveness
- Cumulative reward

7.5 Implementation Process



8. TEST CASES AND EXPECTED / ACTUAL RESULTS

8.1 Test Cases

The system is tested using different simulated patient health conditions to verify whether the RL agent selects suitable actions.

Test Case	Patient State	Expected Action	Actual Action	Result
TC01	Stable	Monitoring	Monitoring	Pass
TC02	Mild	Medication	Medication	Pass
TC03	Moderate	Increased Monitoring	Increased Monitoring	Pass
TC04	Severe	Hospital Referral	Hospital Referral	Pass
TC05	Recovery	Monitoring	Monitoring	Pass

8.2 Test Case Observations

- For a **Stable** patient, the system recommends continued monitoring.
- For a **Mild** condition, the learned policy recommends medication.
- For a **Moderate** condition, increased monitoring is selected.
- For a **Severe** condition, hospital referral is recommended.
- For a patient in **Recovery**, monitoring is continued.

The test results show that the learned policy responds differently to different patient conditions and selects actions according to the learned Q-values.

8.3 Validation

The system is considered successful when the RL agent:

- Selects the expected action for each state.
- Produces positive rewards for improvement and recovery.
- Achieves a higher recovery rate.
- Generates a positive cumulative reward over training episodes.

9. EXECUTION SCREENSHOTS / OUTPUTS

The following screenshots should be included in the report to demonstrate the working of the system.

9.1 Program Execution

Show the Python program running successfully in **VS Code**

Terminal. The screenshot should show:

- Program started successfully
- Training episodes completed
- No errors

9.2 Patient State and Recommended Action

Show an example output such as:

Patient State: Moderate

Recommended Action: Increased Monitoring

Reward: +5

Next State: Mild

9.3 Learned Optimal Policy

Show the final policy generated by the RL agent.

Patient State	Optimal Action
Stable	Monitoring
Mild	Medication
Moderate	Increased Monitoring
Severe	Hospital Referral
Recovery	Monitoring

9.4 Performance Output

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RL-BASED PATIENT HEALTH MONITORING SYSTEM

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Training completed successfully.
Training Episodes: 5000

Learned Q-Table

State	Monitoring	Medication	Increased	Hospital
Stable	20.00	18.00	18.00	18.00
Mild	26.05	20.00	28.95	25.95
Moderate	17.50	28.05	31.05	17.50
Severe	12.50	12.50	17.50	25.00
Recovery	0.00	0.00	0.00	0.00

Optimal Policy

Stable -> Monitoring
Mild -> Increased Monitoring
Moderate -> Increased Monitoring
Severe -> Hospital Referral
Recovery -> Monitoring

Policy Testing

Stable	Action: Monitoring	Recovery: 0/100	Effectiveness: 0/100
Mild	Action: Increased Monitoring	Recovery: 0/100	Effectiveness: 1000/100
Moderate	Action: Increased Monitoring	Recovery: 0/100	Effectiveness: 1000/100
Severe	Action: Hospital Referral	Recovery: 100/100	Effectiveness: 0/100

Performance Results

Intervention Recovery Rate: 33.33%
Treatment Effectiveness: 666.67%
Average Cumulative Reward: 45.77

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FINAL RESULT

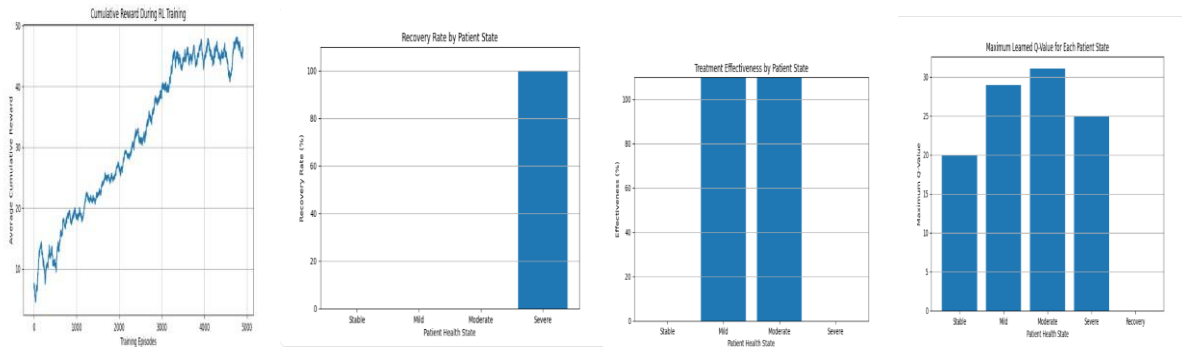
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RL training completed successfully.
Optimal policy generated successfully.
Performance evaluation completed.
Four graphs generated successfully.

9.5 Graphical Output

Include graphs for:

- Cumulative Reward vs Training Episodes
- Recovery Rate
- Treatment Effectiveness



10. RESULTS AND VALIDATION

10.1 Results

After training the RL agent, the system learns an optimal action for each patient health state. The learned policy helps the agent choose actions that can lead to better simulated health outcomes.

10.2 Performance Metrics

The system is evaluated using the following metrics:

Metric	Purpose
Recovery Rate	Measures the percentage of patients reaching the Recovery state

Treatment Effectiveness	Measures how often selected actions improve the patient condition
Cumulative Reward	Measures the total reward obtained during an episode
Policy Accuracy	Measures how consistently the learned policy selects suitable actions

10.3 Learned Policy

Patient State	Learned Action
Stable	Monitoring
Mild	Medication
Moderate	Increased Monitoring
Severe	Hospital Referral
Recovery	Monitoring

10.4 Experimental Observation

The RL agent initially explores different actions and may produce lower rewards. With repeated training, it learns which actions provide better outcomes. The cumulative reward is expected to improve as the agent learns the environment.

10.5 Result Validation

The system is considered effective in the simulation if:

- The learned policy selects suitable actions for different states.
- Patient conditions improve through appropriate actions.
- Recovery rate increases.
- Cumulative reward improves during training.
- The final policy remains consistent during testing.

11. ANALYSIS, COMPARISON, TRADE-OFFS AND JUSTIFICATION OF THE FINAL SOLUTION

11.1 Analysis

The RL-based system learns suitable actions by interacting with the simulated patient environment. Unlike a fixed rule-based system, the RL agent considers the current state, reward, and future outcomes while selecting an action.

The learned policy can adapt its recommendation according to different patient conditions.

11.2 Comparison

Feature	Rule-Based System	RL-Based System
Decision method	Fixed rules	Learned policy
Adaptability	Low	High
Learning	No	Yes
Future outcomes	Limited consideration	Considered
Reward-based optimization	No	Yes
Changing conditions	Less flexible	More flexible

11.3 Trade-

Offs

Advantages:

- Learns from experience.
- Adapts to changing patient states.
- Considers long-term rewards.
- Provides an optimal policy.
- Can be evaluated using measurable performance metrics.

Limitations:

- Requires sufficient training episodes.

- Results depend on the reward design.
- The simulated environment may not represent real clinical situations.
- Incorrect transition assumptions can affect the learned policy.

11.4 Justification of the Final Solution

The RL-based approach is suitable because patient conditions can change over time and decisions may affect future outcomes. MDP and Bellman-based learning allow the system to consider both current rewards and future states.

Therefore, the proposed solution provides a simple and effective demonstration of **Reinforcement Learning for adaptive healthcare decision support**.

12. BROADER CONSIDERATIONS / SDG RELEVANCE

12.1 Broader Considerations

The proposed system demonstrates how Reinforcement Learning can be applied to healthcare decision-support problems. It can help in monitoring changing patient conditions and selecting suitable actions based on learned outcomes.

However, the system is a **simulation** and should not be used for real medical diagnosis or treatment without validation, clinical expertise, and appropriate safety controls.

12.2 SDG Relevance

SDG 3 – Good Health and Well-Being:

The system supports intelligent patient monitoring and aims to improve simulated recovery and timely intervention.

SDG 9 – Industry, Innovation and Infrastructure:

It applies AI, MDP, Bellman equations, and Reinforcement Learning to develop an innovative healthcare-support solution.

SDG 10 – Reduced Inequalities:

A consistent decision-support approach can help provide standardized recommendations across different simulated patient conditions.

12.3 Responsible AI Considerations

The system should consider **patient privacy, transparency, fairness, safety, and human supervision** when adapted for real-world healthcare applications. Final medical decisions should always remain with qualified healthcare professionals.

13. CONCLUSION, LIMITATIONS AND POSSIBLE IMPROVEMENTS

13.1 Conclusion

The proposed system demonstrates the application of **Reinforcement Learning, Markov Decision Process, and Bellman equations** for patient health monitoring and treatment recommendation. The RL agent learns suitable actions for different simulated patient conditions and develops an optimal policy.

The system is evaluated using **recovery rate, treatment effectiveness, and cumulative reward**. The project shows how RL can support adaptive decision-making in healthcare.

13.2 Limitations

- The system uses a simulated patient environment.
- Only a limited number of health states and actions are considered.
- Transition probabilities and rewards are predefined.
- Real patient data is not used.
- The system cannot replace professional medical decision-making.

13.3 Possible Improvements

- Use real and larger healthcare datasets.
- Include more patient health parameters.
- Use advanced RL algorithms such as Deep Q-Networks.
- Add real-time patient monitoring using sensors.
- Develop a web or mobile dashboard.
- Include explainable AI for treatment recommendations.
- Perform extensive validation with healthcare professionals.

14. INDIVIDUAL CONTRIBUTION OF GROUP MEMBERS

Kushitha G – 192425329

B.Tech Artificial Intelligence and Machine Learning

I independently completed the entire assignment on the **RL-Based Patient Health Monitoring and Treatment Recommendation System using Markov Decision Process and Bellman Equations**. I was responsible for understanding and formulating the healthcare problem as a Reinforcement Learning problem and designing the complete MDP framework. I defined the patient health states, available treatment and monitoring actions, reward structure, transition process, and discount factor required for the proposed system.

I developed the problem formulation using the MDP model (S, A, P, R, γ) and applied the Bellman equation to explain how the RL agent can determine the long-term value of different actions. I also identified the objectives, expected outcomes, software and hardware requirements, functional requirements, constraints, and assumptions of the system. I incorporated relevant Reinforcement Learning concepts such as states, actions, rewards, policy, optimal policy, and policy evaluation into the proposed solution.

I designed the complete system workflow, starting from observing the patient's current health condition, selecting an action through the RL agent, receiving a reward, determining the next health state, and updating the learned policy. I also developed the methodology for repeated learning episodes and the generation of an optimal treatment recommendation policy. Finally, I planned the evaluation of the learned policy using **treatment effectiveness, recovery rate, and cumulative reward**. Thus, I was responsible for the complete analysis, formulation, design, methodology, and documentation of this assignment.

15. REFERENCES

1. Richard S. Sutton and Andrew G. Barto, *Reinforcement Learning: An Introduction*, 2nd Edition, MIT Press.
2. Christopher M. Bishop, *Pattern Recognition and Machine Learning*, Springer.
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4. NumPy Documentation, **Numerical Computing with Python**.
5. Pandas Documentation, **Data Analysis and Manipulation with Python**.
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7. OpenAI, **AI and Machine Learning Resources**.
8. World Health Organization (WHO), **Digital Health and Artificial Intelligence in Healthcare**.

9. United Nations, **Sustainable Development Goals – SDG 3: Good Health and Well-Being.**
10. Course materials and lecture notes for **MLA0301 – Reinforcement Learning**, AI & ML Department.

16. ONE-PAGE INDIVIDUAL REFLECTION

Kushitha G – 192425329

B.Tech Artificial Intelligence and Machine Learning

I independently completed the entire project, taking responsibility for all major stages from initial problem identification to final documentation. I formulated the healthcare problem as a Reinforcement Learning problem using a Markov Decision Process (MDP) and defined the patient health conditions as states and treatment or monitoring interventions as actions. I designed the reward function to represent patient outcomes and applied the Bellman equation to explain the process of selecting actions based on immediate and future rewards. I developed the simulated patient environment and defined the state transition process for representing changing patient conditions. I also designed the learning methodology through repeated interactions between the RL agent and the environment to obtain an optimal policy. In addition, I prepared the system workflow, functional requirements, objectives, assumptions, constraints, and expected outcomes. I planned the evaluation of the learned policy using treatment effectiveness, recovery rate, and cumulative reward to measure system performance. I incorporated the relevant Reinforcement Learning concepts throughout the project and organized the complete technical explanation and documentation. Overall, I was responsible for the complete analysis, design, methodology, implementation approach, evaluation strategy, and documentation of the project.